SANAH SURI

Ph.D. in Mathematics
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EDUCATION

Washington University in St. Louis *August 2025*
Ph.D. in Mathematics
Thesis: Functional Equivariance and Backward Error Analysis
Advisor: Dr. Ari Stern

Washington University in St. Louis *May 2022*
A.M. in Mathematics

Grinnell College *May 2020*
B.A. in Mathematics (with honors) & Computer Science

RESEARCH INTERESTS

Machine Learning, Physical Oceanography, Explainable & Interpretable AI, Numerical Analysis

PUBLICATIONS

- **S.Suri**, K. Ringel, and M. Sonnewald (2026) OceanCBM: A Concept Bottleneck Model for Mechanistic Interpretability in Ocean Forecasting, *Submitted to NeurIPS*.
- **S. Suri**, and M. Sonnewald (2026), Trusting Machine Learning with Physics: A Fidelity Verification Framework for Neural Networks, *In Review at Journal of Geophysical Research: Machine Learning and Computation*.
- **S. Suri** (2025), Functional Equivariance and Backward Error Analysis, *Ph.D. Thesis*.
- A. Stern, and **S. Suri** (2023), Functional Equivariance and Modified Vector Fields, *Journal of Computational Dynamics*, 11 (4), 409-426.
- D. Rim, **S. Suri**, S. Hong, K. Lee, R. J. LeVeque (2023), A Stability Analysis of Neural Networks and Its Application to Tsunami Early Warning, *Journal of Geophysical Research: Machine Learning and Computation*, 1 (4).
- M. Chamberland, S. Jing, **S. Suri** (2020), A Generalization of the One-Seventh Ellipse, *Mathematics Magazine*,
93:4, 271-275.
- A. Stern, **S. Suri**, Conjugate Functional Equivariance, In Preparation.

WORK EXPERIENCE

Postdoctoral Scholar Aug 2025-Present
UC Davis, Computational Climate and Ocean Group *Davis, CA*

- Developing interpretable machine learning models for ocean dynamics, focusing on validating learned representations of physical processes
- Mentoring graduate students through reading groups and collaborative research projects in climate AI
- Utilizing high-performance computing (HPC) for large-scale model training and analysis of ocean and climate datasets
- Contributing to grant proposals and interdisciplinary research initiatives in computational climate science

Machine Learning Intern Jun 2022-Aug 2022
Impossible Sensing *St. Louis, MO*

- Developed machine learning models to predict mineral oxide composition from Raman spectra, outperforming baseline methods
- Curated and preprocessed large-scale spectroscopy datasets from open-source mineral databases
- Aligned model development with deep-ocean sensing objectives; presented results to cross-functional teams

PRESENTATIONS

Institute for Pure and Applied Mathematics (IPAM) at UCLA *February 2026*
Poster on 'Trusting Machine Learning with Physics'

UC Davis Coastal and Marine Sciences Institute Symposium *October 2025*
Presentation on 'Trustworthy AI for Ocean Circulation'

WashU Spring Research Roundtable *March 2025*
Presentation on 'Explainable AI in Ocean Science'

WashU Environmental Research Symposium *February 2025*
Selected to present research on 'Explainable AI in Ocean Science'

Missouri S&T Applied Math and Statistics Student Seminar *October 2024*
Invited talk on 'Functional Equivariance and Backward Error Analysis'

SciCADE 2024 *July 2024*
Invited talk in minisymposium 'Recent Advances in Structure Preserving Numerical Methods'

WashU Graduate Research Symposium 2024 *March 2024*
Poster Presentation on 'A Stability Analysis of Neural Networks and Applications to Tsunami Early Warning'

Midwest Numerical Analysis Day *April 2024*
Invited talk in minisymposium 'Numerical Time Integration'

8th Annual Meeting of SIAM Central States Section *October 2023*
Invited talk on 'Modified Vector Fields and Functional Equivariance'

Szego Seminar *March 2023*
Expository talk on 'B-Series' in a graduate student seminar

Joint Math Meetings (JMM) *January 2020*
Poster presentation on 'Colorings of Algebraic Structures'

Nebraska Conference for Undergraduate Women in Math (NCUWM) *January 2019*
Talk on 'The One-Seventh Ellipse Problem'

Joint Math Meetings (JMM) *January 2019*
Poster presentation on 'The One-Seventh Ellipse Problem'

CONFERENCES & WORKSHOPS

ICERM Hot Topics Workshop: From Modeling to Learning with HPC *September 2025*
Workshop on simulation and machine learning integration for scientific computing

Simons Laufer Mathematical Sciences Institute (formerly MSRI) *Summer 2023*
Graduate summer school on machine learning and topological data analysis at UCSD

Midstates Consortium for Math and Science (MCMS) Undergraduate Research Symposium *November 2022*
Graduate student panelist

7th Annual Meeting of SIAM Central States Section *October 2022*
Attended

SIAM Annual Meeting *July 2022*
Attended

Internship Network in the Mathematical Sciences (INMAS) *October 2021-May 2022*
Workshop spanning academic year 2021-22 training PhD students in Mathematics to develop strong programming, statistics, machine learning and data analysis skills

SAMSI Workshop on Data-Driven Mathematical and Statistical Modeling *Summer 2021*
Intensive workshop held annually on statistical and mathematical modeling using real data culminating in a focused project on Bayesian inference

Midstates Consortium for Math and Science (MCMS) Undergraduate Research Symposium *November 2020*
Graduate Student Panelist

TEACHING EXPERIENCE

Assistant to the Instructor May 2021 - May 2024
Washington University in St. Louis *St. Louis, MO*

- Spring 2024: Calculus 3
- Fall 2023: Calculus 3
- Spring 2023: Calculus 1
- Fall 2022: Calculus 1
- Spring 2022: Calculus 3
- Fall 2021: Calculus 3
- Summer 2021: Calculus 2
- Tutor in Mathematics and Statistics Help Room

Directed Reading Program January 2023 - May 2024
Washington University in St. Louis *St. Louis, MO*

- Spring 2024: Mathematical Machine Learning
- Spring 2023: Introduction to B-Series

HONORS & AWARDS

Co-Investigator, NAIRR Pilot Start-Up Project *March 2026*
Second Place, WashU Graduate Research Symposium *March 2024*
Brian Blank Award for Outstanding Third Year Graduate Student *May 2023*
Outstanding Poster at Joint Math Meetings *January 2020*
Third Place, Iowa Intercollegiate Math Competition *Spring 2018*

LEADERSHIP & SERVICE

President and Founder, SIAM Student Chapter at WashU *Fall 2023 - May 2025*
Department Representative, Arts & Sciences Graduate Student Association *August 2023 - May 2024*
Member, Diversity, Equity and Inclusion Committee *August 2022 - May 2023*
Peer Mentor, WashU Department of Mathematics *August 2022 - May 2023*

PROFESSIONAL MEMBERSHIPS

Association for Women in Mathematics (AWM)
American Mathematical Society (AMS)
Society for Industrial and Applied Mathematics (SIAM)